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Abstract

Large vision–language models often hallucinate content unsupported by the image. Training-free contrastive decoding can reduce these errors, but commonly requires multiple model evaluations. ONLY avoids this cost by creating an auxiliary distribution from a TVER-based head mask in one forward process, yet freezes the mask at a single early layer.

We introduce *Adaptive Dual-Branch Decoding* (ADBD), which accumulates continuous TVER evidence over four decoder layers using head-specific adaptive gains. The resulting state forms complementary high- and low-TVER ghost branches. Their distances from the normal distribution independently determine whether each branch is reinforced or penalized in an anchored three-branch rule.

On full POPE with LLaVA-1.5-7B, ADBD reaches 82.10%, 86.37%, and 89.13% accuracy on Random, Popular, and Adversarial, yielding 85.87% macro accuracy and 85.48% macro F1. On MME Hallucination, it improves Position and Color but declines on Existence and Count. The category scores total 630.00 versus 635.55 for ONLY, although the supplied aggregate is 620.00 and requires verification. ADBD remains training-free and avoids a separate full-model invocation, but its gains are category-dependent.

1 Introduction

Large vision–language models (LVLMs) have demonstrated remarkable capabilities in image captioning, visual question answering, and visual reasoning [**?**, **?**, **?**]. Despite this progress, they remain susceptible to hallucination: generating objects, attributes, or relations that are unsupported by the visual input [**?**, **?**]. Such errors reduce user trust and restrict deployment in safety-sensitive settings including medical imaging, autonomous navigation, and assistive technologies.

A prominent family of training-free mitigation methods operates during decoding. Contrastive approaches compare the model's standard next-token distribution with a deliberately distorted counterpart and use their disagreement to reduce reliance on unsupported language priors. Visual Contrastive Decoding (VCD) [**?**], for example, constructs the distorted distribution from a perturbed image, while M3ID [**?**] alters the multimodal conditioning signal. Although

effective, methods that independently evaluate normal and distorted inputs require multiple model passes per generation step, increasing latency and memory consumption.

ONLY [**?**] avoids this duplication by producing a normal distribution and an auxiliary contrastive distribution within one forward process. At a selected intervention layer, ONLY calculates a Text-to-Visual Entropy Ratio (TVER) for every attention head. Heads with low TVER are suppressed to create a textually enhanced hidden representation, which is propagated through the remaining layers using a lightweight ghost path and converted into auxiliary logits. A TVD-based switching rule then either reinforces the auxiliary logits when both paths agree or subtracts them when their distributions diverge.

Despite its efficiency, ONLY makes its head-selection decision from a single layer, typically layer 0. This creates two limitations. First, the decision cannot be revised using deeper representations, even though cross-modal associations evolve throughout the decoder. A head selected from noisy early-layer attention remains selected for the rest of generation, whereas a head that becomes problematic only at a later layer is never captured. Second, a single binary mask collapses a continuous spectrum of text–visual attention behavior into one auxiliary path. It therefore cannot separately represent the complementary behaviors of heads above and below the TVER reference level.

We address these limitations with *Adaptive Dual-Branch Decoding* (ADBD). ADBD accumulates continuous, layer-specific TVER evidence instead of freezing a binary mask. For each head, the method maintains a running state and updates it with a bounded adaptive gain. The gain is head-specific: confident and informative evidence can revise the state more strongly, while weak or redundant observations produce smaller changes. The resulting continuous state partitions the heads into complementary high-TVER and low-TVER groups. Both groups are retained as separate auxiliary branches, producing three next-token distributions in total: the normal distribution, a high-TVER branch, and a low-TVER branch.

The two auxiliary distributions are not forced to follow the same decoding regime. ADBD independently computes their TVD from the normal distribution. Each branch is reinforced when it remains close to the normal path and penalized when it diverges beyond its branch-specific threshold. Whenever a branch is subtracted, the normal logits receive a corresponding anchor increase, preventing auxiliary subtraction from overwhelming the base model. This construction yields four possible joint regimes and allows the two complementary branches to contribute differently at the same generation step.

Our contributions are as follows:

1. We introduce a *continuous multi-layer TVER state* that preserves the magnitude and direction of head-level evidence instead of immediately reducing every layer to a binary mask.

2. We propose a *head-specific adaptive update mechanism* that controls how strongly new layer evidence changes the accumulated state using bounded gains derived from confidence and innovation.

3. We construct *complementary high-TVER and low-TVER ghost branches* and combine them with the normal distribution through an independently switched, anchored three-branch decoding rule.

4. We evaluate the method on POPE and the MME Hallucination subset using LLaVA-1.5-7B. On the full POPE benchmark, ADBD reaches 82.10%, 86.37%, and 89.13% accuracy on Random, Popular, and Adversarial, respectively, with 85.87% macro accuracy and 85.48% macro F1. MME provides a complementary diagnosis of category-specific gains and regressions. Component variants are reported only as ablations.

The remainder of this paper is organized as follows. Section **??** reviews related work, Section **??** presents ADBD, Section **??** describes the experimental evaluation, and Section **??** concludes the paper.

2 Related Work

Complete the Related Work section. It is intentionally omitted from the current rewrite.

3 Method

We present *Adaptive Dual-Branch Decoding* (ADBD), a training-free decoding method that augments the normal LVLM path with two complementary auxiliary paths. ADBD consists of four stages: computing layer-specific TVER evidence, accumulating this evidence through per-head adaptive updates, constructing complementary high- and low-TVER ghost branches, and resolving the three distributions with independently switched, anchored decoding.

3.1 Preliminaries: ONLY

ONLY [**?**] produces a normal distribution and one auxiliary contrastive distribution in a single forward process. We briefly review the components retained by ADBD.

3.1.1 Text-to-Visual Entropy Ratio

At decoder layer ℓ, let $a_{\ell,h}$ denote the attention weights of head h for the final query position. The attention vector is divided into textual and visual subsets according to token position:

$$a^{\mathcal{T}}_{\ell,h} = \{a_{\ell,h,j} \mid j \in \mathcal{I}_{\mathcal{T}}\}, a^{\mathcal{V}}_{\ell,h} = \{a_{\ell,h,j} \mid j \in \mathcal{I}_{\mathcal{V}}\}. \tag{1}$$

Following ONLY, values exceeding the subset mean plus one standard deviation are removed and each subset is renormalized. The textual and visual entropies

are

$$H^{\mathcal{T}}_{\ell,h} = -\sum_{j \in \mathcal{I}_{\mathcal{T}}} \hat{a}^{\mathcal{T}}_{\ell,h,j} \log\big(\hat{a}^{\mathcal{T}}_{\ell,h,j} + \epsilon\big)\,, \tag{2}$$

$$H^{\mathcal{V}}_{\ell,h} = -\sum_{j \in \mathcal{I}_{\mathcal{V}}} \hat{a}^{\mathcal{V}}_{\ell,h,j} \log\big(\hat{a}^{\mathcal{V}}_{\ell,h,j} + \epsilon\big)\,. \tag{3}$$

The TVER of head h is

$$r_{\ell,h} = \frac{H^{\mathcal{T}}_{\ell,h}}{H^{\mathcal{V}}_{\ell,h}}. \tag{4}$$

A low ratio indicates concentrated textual attention relative to diffuse visual attention, whereas a high ratio indicates the complementary entropy pattern.

3.1.2 One-Layer Mask and Ghost Path

ONLY computes a binary mask at one intervention layer $\tilde{\ell}$ using the mean ratio across heads:

$$m_{\tilde{\ell},h} = 1\left[r_{\tilde{\ell},h} \geq \frac{1}{H}\sum_{j=1}^{H} r_{\tilde{\ell},j}\right]. \tag{5}$$

The attention outputs of suppressed heads are zeroed to create an auxiliary hidden state. This state is threaded through the remaining decoder layers without the complete normal residual and feed-forward computation, then processed and anchored to the normal representation at the final layer. Applying the language-model head produces the auxiliary logits.

3.1.3 Single-Branch TVD Switching

Let f^0_t and f^{aux}_t be the normal and auxiliary logits at autoregressive step t, with probability distributions p^0_t and p^{aux}_t. ONLY measures

$$d_t = \frac{1}{2}\sum_{y \in \mathcal{V}} \left|p^0_t(y) - p^{\text{aux}}_t(y)\right|. \tag{6}$$

The auxiliary logits are added when $d_t < \gamma$ and subtracted otherwise. A plausibility constraint [**?**] removes candidates whose normal logits fall below

$$\log\beta + \max_{y \in \mathcal{V}} f^0_t(y). \tag{7}$$

3.2 Adaptive Dual-Branch Decoding

3.2.1 Continuous Layer-Wise TVER Evidence

A binary mask records only whether a head lies above or below a threshold. It discards the distance from that threshold and cannot express uncertainty near

the boundary. ADBD therefore maps the raw TVER vector at every selected layer to continuous evidence. Let

$$\bar{r}_\ell = \frac{1}{H}\sum_{h=1}^{H} r_{\ell,h}, \qquad \sigma_\ell = \sqrt{\frac{1}{H}\sum_{h=1}^{H}(r_{\ell,h} - \bar{r}_\ell)^2}. \tag{8}$$

The frozen implementation defines

$$e_{\ell,h} = sigmoid\left(\frac{r_{\ell,h} - \bar{r}_\ell}{\max(\sigma_\ell, 10^{-6})}\right). \tag{9}$$

Thus, $e_{\ell,h} \in (0,1)$, values above 0.5 denote the high-TVER side of the layer reference, and the distance from 0.5 retains the strength of that evidence.

3.2.2 Per-Head Adaptive State Update

For each head index h, ADBD maintains a continuous high-TVER state $s_{\ell,h}$. At the first selected layer ℓ_0, the state is initialized as

$$s_{\ell_0,h} = e_{\ell_0,h}. \tag{10}$$

At every later selected layer, the implementation computes confidence and innovation as

$$c_{\ell,h} = |2e_{\ell,h} - 1|, \qquad \iota_{\ell,h} = \left|e_{\ell,h} - s_{\ell^-,h}\right|, \tag{11}$$

where ℓ^- is the previous selected layer. The head-specific gain is

$$g_{\ell,h} = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) c_{\ell,h}\iota_{\ell,h}, \tag{12}$$

and the state update is

$$s_{\ell,h} = (1 - g_{\ell,h}) s_{\ell^-,h} + g_{\ell,h} e_{\ell,h}. \tag{13}$$

Because both confidence and innovation lie in $[0,1]$, the gain is bounded by $\alpha_{\min} \le g_{\ell,h} \le \alpha_{\max}$. Strong evidence that differs from the accumulated state receives a larger update, whereas evidence near the layer mean or close to the current state produces a smaller update. At the first selected layer, the state is initialized directly and the logged gain is zero.

The update is performed only at a chosen set of layers $\mathcal{L}_{\text{upd}}$. Non-selected layers reuse the most recent state. In the default LLaVA-1.5-7B configuration, layers 0–30 are eligible for accumulation and layer 31 is reserved for final auxiliary-path integration.

3.2.3 Complementary Head Partition

The implementation stores the complementary state $1 - s_{\ell,h}$ alongside $s_{\ell,h}$. The accumulated state partitions the heads into disjoint and exhaustive sets:

$$k^+_{\ell,h} = 1[s_{\ell,h} \ge 0.5], \qquad k^-_{\ell,h} = 1 - k^+_{\ell,h}. \tag{14}$$

The + mask retains heads whose accumulated evidence lies on the high-TVER side, while the − mask retains heads on the low-TVER side. We use the descriptive names *high-TVER branch* and *low-TVER branch*; these avoid asserting an intrinsic visual or textual specialization that has not been independently established.

For branch $b \in \{+, -\}$, masked attention is

$$\tilde{a}^b_{\ell,h} = \{\, a_{\ell,h}, k^b_{\ell,h} = 1, 0, k^b_{\ell,h} = 0. \tag{15}$$

This produces two auxiliary attention outputs at each relevant layer.

3.2.4 Dual Ghost Paths

Let h^0_ℓ denote the normal hidden representation and h^+_ℓ, h^-_ℓ denote the two auxiliary representations. Both auxiliary states are propagated through the decoder's ghost-path mechanism. At the final layer, each branch is normalized, receives a residual anchor from the normal path, passes through the final MLP, and is again anchored to the normal final representation before the language-model head. This yields three logit vectors:

$$f^0_t, \qquad f^+_t, \qquad f^-_t. \tag{16}$$

Both auxiliary branches share the same integration structure; their distinction arises solely from the complementary head masks.

3.2.5 Independent Branch Disagreement

ADBD calculates branch-specific TVDs from the normal distribution: $\mathrm{d}^+_t = \frac{1}{2}\sum_{y\in\mathcal{V}} \left|p^0_t(y) - p^+_t(y)\right|$,
$d^-_t = \frac{1}{2}\sum_{y\in\mathcal{V}} \left|p^0_t(y) - p^-_t(y)\right|$. Each distance is compared with its own threshold γ_+ or γ_-. Consequently, one branch can be collaborative while the other is contrastive at the same generation step.

3.2.6 Anchored Three-Branch Resolution

The high-TVER branch contributes

$$\Delta^+_t = \{\, \alpha^{\mathrm{col}}_+ f^+_t, d^+_t < \gamma_+, \; -\alpha^{\mathrm{con}}_+ f^+_t, d^+_t \geq \gamma_+, \tag{17}$$

and the low-TVER branch contributes

$$\Delta^-_t = \{\, \alpha^{\mathrm{col}}_- f^-_t, d^-_t < \gamma_-, \; -\alpha^{\mathrm{con}}_- f^-_t, d^-_t \geq \gamma_-. \tag{18}$$

To stabilize subtraction, ADBD increases the normal-logit anchor for every branch that enters the contrastive regime:

$$\kappa_t = 1 + \alpha^{\mathrm{con}}_+ 1[d^+_t \geq \gamma_+] + \alpha^{\mathrm{con}}_- 1[d^-_t \geq \gamma_-]. \tag{19}$$

The final logits are

$$f^{\mathrm{final}}_t = \kappa_t f^0_t + \Delta^+_t + \Delta^-_t. \tag{20}$$

This rule yields four joint regimes: both auxiliary distributions may be reinforced, either one may be reinforced while the other is penalized, or both may be penalized while the normal anchor is strengthened twice. The adaptive plausibility constraint is applied using the normal logits before sampling.

3.3 Algorithm Summary

1. Initialize the continuous head state $s \leftarrow \emptyset$.
2. For each decoder layer ℓ, compute the normal self-attention output.
3. If $\ell \in \mathcal{L}_{\text{upd}}$, compute TVER, map it to continuous evidence with Eq. eq:continuous$_evidence, andinitialized$
$-eq : state_update.$
3. Construct the complementary masks in Eq. eq:complementary$_masks, thenproduceandpropagatethetwoma$
3. At the final layer, integrate both ghost states and obtain f_t^+ and f_t^- alongside f_t^0.
4. Compute d_t^+ and d_t^-, independently select the collaborative or contrastive contribution of each branch, and form the anchored logits in Eq. eq:final$_logits$.
4. Apply the adaptive plausibility constraint and sample the next token.

3.4 Computational Characteristics

ADBD is training-free and avoids a second full-model invocation. It reuses normal attention tensors for two masked outputs and propagates two auxiliary states. Relative to ONLY, it adds a second branch, per-layer TVER updates, and one language-model-head projection; their runtime and memory costs require empirical measurement.

4 Experiments

We evaluate ADBD on POPE [**?**] and the hallucination subset of MME [**?**]. Earlier proposal variants appear only as ablations.

4.1 Experimental Setup

Model. We use LLaVA-1.5-7B [**?**] for all experiments.

POPE. POPE uses balanced yes–no questions from COCO images [**?**] with Random, Popular, and Adversarial negative sampling. We evaluate all 3,000 questions in each split.

Split	Acc.	Prec.	Rec.	F1
Random	82.10	81.78	82.60	82.19
Popular	86.37	89.16	82.80	85.86
Adversarial	89.13	94.81	82.80	88.40
Macro	85.87	–	–	85.48

Table 1: ADBD results on full POPE splits (%).

Method	Exist.	Count	Pos.	Color	Total
ONLY	**191.67**	**145.55**	136.66	161.66	**635.55**
ADBD	180.00	121.67	**153.33**	**175.00**	620.00†

Table 2: MME Hallucination scores.

MME Hallucination subset. We use MME's Existence, Count, Position, and Color categories. Add evaluation details and verify the ADBD aggregate: its category scores sum to 630.00, but the supplied total is 620.00.

Baselines and ablations. We compare ADBD with standard LLaVA and ONLY [**?**]. Earlier variants are used only for component ablations.

Implementation details. ADBD accumulates over four layers and generates up to eight tokens. Its branch coefficients are $(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = (0.4, 1.8, 0.35, 2.2)$, with both TVD thresholds set to 0.15. Report the four layer indices, gain bounds, β, sampling parameters, seed, and hardware.

Metrics. For POPE, we report Accuracy, Precision, Recall, and F1 for each split, plus macro Accuracy and F1. For MME, we report four category scores and their total.

4.2 POPE Main Results

Table **??** reports ADBD on the full POPE splits.

ADBD obtains 85.87% macro accuracy and 85.48% macro F1. Accuracy rises from 82.10% on Random to 89.13% on Adversarial. Recall stays near 82.7%, while precision rises from 81.78% to 94.81%, indicating fewer false positives on harder negative samples. Matching full-split LLaVA and ONLY results are needed for a controlled baseline comparison.

4.3 MME Hallucination Results

Table **??** reports the four MME categories.

†The displayed ADBD category scores sum to 630.00, which would imply a difference of -5.55 rather than -15.55. The experiment log or evaluation output must be checked before submission.

Variant	Multi	Cont.	Gain	Dual	Switch
ONLY	N	N	N	N	N
Binary accum.	Y	N	N	N	N
Continuous accum.	Y	Y	N	N	N
Static dual	N	N	N	Y	Y
Global EMA	Y	N	N	Y	Y
ADBD	Y	Y	Y	Y	Y

Table 3: Ablation design.

ADBD improves Position by 16.67 points and Color by 13.34, but decreases Existence by 11.67 and Count by 23.88. It therefore does not improve the MME total over ONLY, showing that its gains are category-dependent.

4.4 Ablation Study

We use earlier proposal variants only as ablations that remove or simplify ADBD components. Table **??** summarizes their structure.

Add controlled ablation results when available.

4.5 Limitations

ADBD has several limitations. First, evaluation on POPE and the MME Hallucination subset shows that the method does not improve every visual competency uniformly. Although ADBD improves MME Position and Color, it underperforms ONLY on Existence, Count, and the reported aggregate score. POPE and MME also rely primarily on constrained yes–no evaluation and do not establish generalization to open-ended captioning. Evaluation on CHAIR [**?**] and additional hallucination benchmarks is therefore needed.

Second, the implementation uses fixed textual and visual token boundaries tailored to the evaluated LLaVA-1.5 prompt and its 576 image tokens. This assumption may fail under another conversation template, visual resolution, processor, or multi-image input. A robust implementation should obtain modality boundaries from the tokenizer and multimodal processor.

Third, accumulating a state at the same head index across layers implicitly treats those coordinates as comparable. Attention head h in one layer is not guaranteed to have the same function as head h in another. ADBD should therefore be interpreted as accumulating evidence over aligned mask coordinates rather than tracking a semantically identical head through depth. Future work could use layer-specific statistics or estimate cross-layer correspondences.

Fourth, ADBD introduces computation and memory relative to ONLY because it maintains two auxiliary states. It remains training-free and avoids a separate full-model pass, but its practical efficiency depends on measured overhead.

Fifth, TVER-based branch names do not prove that one branch is intrinsically visual and the other intrinsically textual. Attention visualization, controlled modality ablations, and branch-removal experiments are required before making stronger semantic claims.

Finally, if results are reported from a subset or a single stochastic seed, the statistical strength of the conclusions is limited. The final evaluation should use the complete benchmark or a pre-specified sampling protocol and report variability across seeds where sampling-based decoding is used.

5 Conclusion

We introduced Adaptive Dual-Branch Decoding, a training-free method for reducing hallucinations in LVLMs. ADBD replaces ONLY's frozen one-layer binary decision with a continuous TVER state refined across selected decoder layers by head-specific adaptive gains. The state produces complementary high- and low-TVER auxiliary branches, which are independently reinforced or penalized according to their disagreement with the normal distribution. An anchored three-branch rule combines these signals without requiring a separate full LVLM invocation.

On the full POPE benchmark with LLaVA-1.5-7B, ADBD achieves 82.10%, 86.37%, and 89.13% accuracy on Random, Popular, and Adversarial, respectively, yielding 85.87% macro accuracy and 85.48% macro F1. Precision increases across the three splits while recall remains stable, producing the strongest result on Adversarial negatives. On MME Hallucination, the method improves Position and Color but decreases Existence and Count, yielding a reported aggregate score of 620.00 compared with 635.55 for ONLY; the aggregate must be verified because the supplied ADBD category scores sum to 630.00. Component ablations and branch-level decoding analysis remain future experimental work. Overall, the results suggest that complementary and adaptively accumulated attention evidence can provide a useful contrastive signal, but its gains are category-dependent rather than universal.

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